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324871 (24)

BE (8th Semester)

Examination, April - May, 2012

Branch : Electrical, EEE

EHV AC & DC TRANSMISSION

(PE-III)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each question is compulsory which carries two marks. Solve any two parts from part (b), (c) & (d) from each question which carries 7 marks each.

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Q. 1. (a) What are the different types of links used HVDC transmission.

(b) Compare the EHV AC-DC Transmission o the basis of technical performance.

(c) Explain thyristor characteristics in differen states used at converter station. 1

(d) Explain the conduction of thyristors withou overlapping in GARENTZ circuit. 1

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UNIT - II

Q. 2. (a) What do you mean by shunt reactors ? 2

(b) Enlist the different FACTS controllers. Explain the methods used for the compansation of reactive power. 7

(c) Explain the causes of reactive power absorbed by HVDC converter substation. 7

(d) Calculate the reactive power absorbed by rectifier, if : 7

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(i) No load ideal DC voltage equal to 276.75 KV; The DC current is 1.5 KA the actual DC voltage is 250 KV.

(ii) Actual DC voltage 246.25 KV no load DC voltage is 275.53 KV & DC current is 2 KA.

UNIT - III

Q. 3. (a) Differentiate between a hot lightning stroke and a cold lightning stroke. 2

(b) Discuss the causes of over voltages held in the HVDC transmission system. 7

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(c) Explain the expulsion gap type surge arresters. 7

(d) What is an travelling wave ? Explain the development of such a wave on an overhead line. 7

UNIT - IV

Q. 4. (a) Define MTDC System. 2

(b) Discuss the overvoltage protection in a converter station in EHV DC system. 7

(c) Write down the different type of faults occurs in the converters. 7

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(d) What are the main objectives in designing the size and branches of AC harmonic filters and DC harmonic filters in a HVDC substation. 1

UNIT - V

Q. 5. (a) What is a close loop (feed-back) current control system ? 1

(b) Explain parallel operation of HVAC and DC system. Write down their advantages. 1

(c) Describe the individual phase control firing scheme for the operation of constant current controller. 1

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(d) Explain by means of a schematic diagram and equations, how power flow through an HVDC link is controlled. 7

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